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Biochemical and microbial changes reveal how aerobic pre-treatment impacts anaerobic biodegradability of food waste

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ABSTRACT

Aerobic pre-treatment of food waste (FW) was performed at different oxygen concentrations (0%, 5%, 10% and 21%O_2) and different durations (1, 2, 3 and 4 days) to investigate its impact on biochemical and microbial community characteristics of the waste and its ability to improve anaerobic biodegradability. Whatever the duration, the highest effect of pre-treatment was observed at full aerobic pre-treatment (21%O_2) while 5%O_2 and 10%O_2 showed lower transformation performances. Biochemical variations at 21%O_2 were mainly a decrease of simple carbohydrates, volatile fatty acids (VFA) and low molecular weight water soluble compounds and an increase of high weight water soluble compounds. Microbial community analysis showed a clear modification of populations after 21%O_2 aerobic pre-treatment, changing from an initial dominance of lactic acid bacteria to a final dominance of VFA consumers (like *Acetobacter*) and a higher presence of *Fungi*. Enzymatic tests showed an increase of exoenzymes content and a higher presence of protein and carbohydrates degrading enzymes. Finally, the aerobic pre-treatment did not negatively impact methane potential of FW (496 $NLCH_4 \cdot kgVS^{-1}$) which remained unchanged after two days of pre-treatment at 21%O_2. These latter optimal pre-treatment conditions are proposed to be tested in future investigation of anaerobic digestion (AD) process with low inoculum to substrate ratio in order to assess their ability to avoid acidification risk during AD of FW.

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1. Introduction

The management and treatment of food waste (FW) is a subject of growing interest worldwide. In Europe, about 89 Mt of FW are produced each year, mainly by households and food services (53% and 12% of the total FW respectively) (Monier et al., 2010; Stenmarck et al., 2016). Beyond waste prevention, EU is thus promoting FW valorisation, essentially via composting or anaerobic digestion (article 5 of EU Directive 99/31/EC).

Anaerobic digestion (AD) has proven to be an appropriate technique to treat FW, resulting in the production of an energy-rich biogas and a digestate which can be used as biological fertiliser (Li et al., 2011; Zamanzadeh et al., 2016). However, the stability and efficiency of AD of FW can be problematic on two points. On short-term, a rapid accumulation of volatile fatty acids (VFA), caused by the high concentration of simple carbohydrates in FW, may inhibit methane production (Ren et al., 2018; Scano et al., 2014; Zhang et al., 2012). A total inhibition can be observed at a VFA concentration of 40–50 $gCOD \cdot L^{-1}$ (Veeken and Hamelers, 2000). On longer term, anaerobic conditions result in slow hydrolysis kinetics and only partial degradation of the structural carbohydrates of FW: only about 50% of the cellulose is degraded after 30 days of AD (Deublein and Steinhauser, 2011). Moreover, fermentation during storage step between the collection of the FW and its introduction in the AD treatment can produce an additional VFA accumulation contributing to instability (Fisgativa, 2016).

Several studies have been conducted to improve the AD yield and stability, using mechanical, thermal, chemical or enzymatic pre-treatments of the substrates (Brémond et al., 2018; Carrere et al., 2016; Kondusamy and Kalamdhad, 2014). Even though some pre-treatments demonstrated their ability to enhance AD, side effects as the release of recalcitrant and inhibitory molecules or further difficulty for digestate dewatering were also observed. These undesired phenomena, in addition to the difficulty of implementation and high energy requirements of pre-treatments, often counteracted their positive effects (Carlsson et al., 2012; Ma et al., 2011). In comparison, the interest of a biological aerobic pre-treatment of substrates before AD has been revealed more recently with a real potential. Indeed, it has been shown that pre-treating FW enhances the hydrolysis of slowly biodegradable compounds by promoting the microbial excretion of hydrolytic enzymes (Charles et al., 2009; Lim and Wang, 2013; Meng et al., 2016; Sarkar and Venkata Mohan, 2017). Ni et al. (2017) stated

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an increase of methane production during municipal solid waste (MSW) dry anaerobic digestion with an 8-day sequencing pre-aeration because of accelerated degradation of proteins (airflow rate of 0.5 $L \cdot min^{-1} \cdot kg\ waste^{-1}$ performed by cycle of 10 min aeration followed by a 20 min pause). This positive effect of aeration was also observed by Rafieenia et al. (2017) on methane production with one day pre-aerated (5 $L \cdot h^{-1}$) protein-rich FW. However, the duration and conditions of aerobic pre-treatment should be well managed to not oxidise excessively the organic matter of FW and limiting the methane yield. Brummeler and Koster (1990) observed that 10-days of aerobic pre-treatment, on the organic fraction of municipal solid waste (OFMSW), resulted in lower initial acid formation during AD linked to the degradation of the easily degradable part of the substrate but induced a loss of 40% of methane production. Wu et al. (2018) showed that an aerobic storage of FW during 8 days produced a diminution of 15% of the methane yield comparing to raw FW. Charles et al. (2009) showed that carbon emitted as CO_2 (during six days of pressuring aerobic pre-treatment of OFMSW), accounts for the reduction of 26% of the CH_4 produced during subsequent AD. Finally, Rafieenia et al. (2017) observed that even a short (one day) aerobic pre-treatment causes a decrease of methane potential of a lipid-rich FW, compared to a non-pre-aerated sample.

In spite of these studies, there is still a lack of knowledge on the biochemical and microbial community modifications promoted by such aerobic pre-treatments while there is a real challenge at the industrial level in assessing the ability of aerobic pre-treatment to limit the risk of instabilities potentially occurring during AD of FW. The aim of this study was thus (1) to understand how short time aerobic pre-treatments influence further anaerobic biodegradability of the waste and which variations of FW characteristics (physicochemical characteristics, biochemical fractionation, microbial populations and enzymes content), promoted by the pre-aeration explain this influence; and (2) to highlight which pre-treatment condition could be interestingly later tested in AD process conditions. The impacts of the oxygen level and of the pre-treatment duration were specifically assessed. Compared to previously published experiments that were performed at small laboratory scale (1-L bottles) with grounded FW (Meng et al., 2016; Rafieenia et al., 2017; Wu et al., 2018), our study was realised at a 10-L laboratory pilot scale with raw FW.

2. Materials and methods

2.1. Composition and sampling of food waste feedstock

The study used FW (kitchen waste and leftovers of the dishes served) produced by a collective catering establishment, which produces 17 tons of waste per year, corresponding to nearly 700 customers per day. FW was collected over a period of three days (stored at 4 °C) and mixed to obtain a representative sample with the following typological composition: 30% vegetables, 24% fruits, 19% meat, 13% starchy food, 9% bones and shells and 5% paper per wet weight (WW). Initial FW (hereafter referred to as T0 or raw FW) was sub-sampled and stored at −20 °C for later use in the experiments to ensure the same initial feedstock conditions. The characteristics of the raw FW (analysed as described in Sections 2.4 and 2.5 below) are listed in Table 1.

Table 1
Characteristics of the raw food waste.

Parameter	Raw FW (T0)
pH	5.3
Total solids (TS, $g \cdot kgWW^{-1}$)	295.3 ± 14.7
Volatile solids (VS, $g \cdot kgWW^{-1}$)	256.5 ± 13.6
Volatile solids (VS, $g \cdot kgTS^{-1}$)	868.9 ± 20.9
Total chemical oxygen demand (COD, $gO_2 \cdot kgVS^{-1}$)	1714.0 ± 51.4
Total carbon (C, %VS)	56.1 ± 4.0
Total nitrogen (N, %VS)	3.7 ± 0.4
C/N ratio (%)	15.1
Carbohydrate (%VS)	68.8
Protein (%VS)	25.0
Fat (%VS)	6.2
Volatile fatty acids (VFA, $g \cdot kgVS^{-1}$)	9.8
Ammonium content (NH_4^+, $g \cdot kgVS^{-1}$)	0.3 ± 0.01
Biomethane potential (BMP, $NLCH_4 \cdot kgVS^{-1}$)	496.0 ± 31.9

where WW- wet weight.

2.2. Experimental pre-treatment set-up

The pre-treatment pilot consisted of four 10-L glass cells equipped with a gas flow inlet and outlet. The cells containing 3 kg of FW were maintained in a water bath heated by an automated thermostat at a temperature of 37 °C. This temperature is close to conditions of mesophilic AD processes. The temperature within the cell contents was recorded using a Platinum resistance thermometer. The pre-treatment tests were carried out by circulating 50 $L \cdot h^{-1}$ of gas, to optimise the aerobic degradation of FW as described by Tremier et al. (2005), with a recirculation rate of 300 $L \cdot h^{-1}$ to ensure homogeneous conditions. The inlet of gas was located at the bottom of the cell and the gas flowed through the waste to reach the outlet located at the top of cell. The O_2 content in the gas flow entering the experimental cell was set at 0%, 5%, 10% or 21% by mixing compressed air with nitrogen (N_2) to study the effects of anaerobic conditions (0%O_2), limited oxygen conditions (5%O_2 and 10%O_2) and ambient air or full aeration conditions (21%O_2). Incoming and outgoing O_2, CO_2, N_2O and CH_4 concentrations in the air of each cell were measured at 3-min intervals using an ABB gas analyser (model URAS26EL3020, ABB).

Because it is not possible to get a homogeneous FW sample from a cell without disturbing waste characteristics, doing a time course experiment required running as many cells as the number of samples needed. Thus, for each O_2 concentration, four cells were started at the same time and one cell was stopped every day of the 4-day experiment in order to collect and sample the waste and to measure variations in FW characteristics. Along this paper, the durations of 1, 2, 3 and 4 days of pre-treatment, regardless the oxygen conditions, will be also referred as T1, T2, T3 and T4 respectively. The maximal duration of 4 days of pre-treatment was chosen to limit excessive organic matter degradation in aerobic conditions according to previous studies (Charles et al., 2009). The total content of each cell, thereafter called treated FW, was immediately homogenised and frozen at −20 °C prior to analyses.

2.3. Samples preparation

One kilogram of each frozen sample (raw FW and treated FW from each cell) was ground to 2 mm using a Robot Coupe R15 Food Processor for physicochemical and biochemical analyses. About 7 g of these ground FW samples were further ground in liquid nitrogen for 1 min with a ball mill (300 Dangoumill, Prolabo) to obtain a homogeneous powder, which was stored at −80 °C for further molecular microbiology and enzymatic analyses.

2.4. Physicochemical and biochemical analyses

Total solids (TS) were analysed after drying at 80 °C until constant weight. Volatile solids (VS), chemical oxygen demand (COD) total Kjeldahl Nitrogen (TKN) and ammonium (NH_4^+) were measured using standard methods (APHA, 2012).

Aqueous extraction was performed to separate water soluble (SOL_W) and non-water-soluble (NSOL) organic matter. The NSOL

fraction was dried at 80 °C until constant weight and then analysed using the modified Van Soest fibre analysis method, as described by Tremier et al. (2014). This method discretises the organic matter (OM) from NSOL into OM soluble in neutral detergent (SOL_{NDF}), hemicellulose-like OM (HEM), cellulose-like OM (CEL) and lignin-like OM (LIG) fractions. The composition of the organic macromolecules was determined on the SOL_W by a size-exclusion HPLC method as described by Tremier et al. (2014). For the purpose of the present study, water soluble OM molecules bigger and smaller than 1.5 kDa ($SOL_W > 1.5$ and $SOL_W < 1.5$ respectively) were distinguished. This distribution is based on the hypothesis that $SOL_W < 1.5$ molecules can diffuse through the membrane of microorganisms and be used for microbial metabolism (Tremier et al., 2014), while $SOL_W > 1.5$ molecules require prior hydrolysis by exoenzymes before being consumed by microorganisms.

Biochemical fractionation of lipids, proteins and carbohydrates was also performed. To determinate lipids content, a nuclear magnetic resonance (NMR) measurement was performed as described by Picard et al. (2014). The analysis was carried out on the solids fraction after dried at 38 °C until constant weight and grounded at 1.5 mm. In brief, lipids measurements were performed with a low field NMR operating at a frequency of 10 MHz using a Brucker spectrometer Minispec MQ 10. About 1.5 g of dried sample was placed in a 30 mm-diameter NMR tube with an approximate height of 10 mm. Lipids was then calculated on the basis of a simple linear regression (Toussaint et al., 2002). Protein content was calculated using the nitrogen content as described by Dintzis et al. (1988):

$$\text{Protein} = \left(\text{TKN} - \mathbf{NH_4^+}\right) * 6.25$$

where Protein, TKN and NH_4^+ are expressed in g N·kgWW^{-1} and 6.25 is the average conversion factor to estimate the protein content based on a N analysis. Then, carbohydrates content was considered being the residual fraction of VS outwards the sum of lipids and proteins. In addition, VFA (C2 to C5: acetic, propionic, butyric, isobutyric and isovaleric fatty acids) were determined from the SOL_W fraction by a HPLC method using an UV detector, as described by Peu et al. (2004). Simple carbohydrates were determined as the sum of glucose, fructose and sucrose contents using an ELSD detector, as described by Muir et al. (2009).

2.5. Anaerobic biodegradability tests

To determine the methane yield of raw and treated FW, the biological methane potential (BMP) tests were performed according to NF EN ISO 11734 standard as described by Vedrenne et al. (2008). Measurements were made in triplicate using hermetically closed 572 ml-bottles, containing 40 ml of inoculum and 3–5 g of ground FW (raw or treated) (ratio 1 gVS inoculum:1 gVS FW as recommended by Parra-Orobio et al. (2018) and Hobbs et al. (2018)). Measurements were performed on raw FW (T0) and at 2 and 4 days of pre-treatment for each oxygen condition. Samples from 1 and 3 days pre-treatment were not analysed because of limitation on experimental schedule. The inoculum was obtained from a well-established anaerobic pilot digester (87 L) acclimated to degrade pig slurry supplemented with horse feed. The bottles were incubated at 38 °C for about 40 days. Daily pressure measurements enabled quantification of biogas production. Biogas was then sampled for CH_4 and CO_2 analysis by gas chromatography equipped with an electron capture detector (Agilent Technologies, 6890N, USA) according to the method described by Lucas et al. (2007).

2.6. Characterisation and dynamics of FW microbial community

The microbial community of each sample was analysed as described in detail by Fisgativa et al. (2017). Briefly, total DNA was extracted from about 200 mg of frozen ground FW using the Macherey Nagel Genomic DNA for Food kit. Bacterial and fungal communities of samples were quantified by SSU rRNA-targeted real-time qPCR using generic primers. Bacterial and archaeal diversity was determined by high throughput DNA sequencing of the 16S rRNA gene V4-V5 variable regions using Ion Torrent™ PGM technology (Life Technologies, USA) at the IRSTEA BIOMIC team (Antony, France) (Poirier et al. (2016). Between 20,796 and 83,525 DNA sequences were obtained per sample that were processed and clustered into operational taxonomic units (OTUs, based on 97% similarity) using the QIIME software. Taxonomic identification of the OTUs was done on the Ribosomal Database Project database (https://rdp.cme.msu.edu/). OTUs containing less than 1% of the total number of sequences of all samples were considered as subdominant. Fungal diversity was investigated by collecting mycelium on day 4 of the 21%O_2 pre-treatment condition, extracting DNA and cloning and sequencing of a 422 bp fragment of the 18S rRNA genes. The sequences were aligned against the NCBI database for identification using Blast software (NCBI, http://www.ncbi.nlm.nih.gov/BLAST).

2.7. Determination of enzymes

2.7.1. Total hydrolytic exoenzymes activity

The extracellular enzymes (exoenzymes) present in the sample hydrolyse the fluorescein diacetate (FDA) into fluorescein, a soluble molecule which can be quantified by fluorescence (Emission 530). The method used in this study was adapted from Adam and Duncan (2001). The reagents used (potassium phosphate buffer (60 mM, pH 7.6); (b) FDA stock solution (stored at −20 °C); and (c) Pure fluorescein (Sigma-Aldrich Co. Ltd) stock solution (stored at 4 °C in the dark)) were prepared as described by Adam and Duncan (2001).

For the analyses, 5 g of frozen ground FW were mixed with 50 ml sterile water and vigorously stirred with a stomacher (Stomacher 400, Tekmar, Cincinnati, OH, USA) for 10 min. After 10 min of decantation, centrifugation was performed at 3000*g* at 4 °C for 10 min. Next, 2 ml of supernatant were mixed with 15 ml of potassium phosphate buffer solution and 0.2 ml of FDA solution in a 40 ml flask. The flask was closed, mixed by hand and placed on an orbital incubator (150 rpm) at 37 °C for 30 min. Control samples were prepared without the addition of FDA. After incubation, 15 ml of acetone were added immediately to stop the reaction. After 10 min of decantation, the content of each flask was centrifuged (3000*g* at 4 °C for 3 min). The supernatant was filtered (Whatman, No.2) and analysed in a black microwell plate with a fluorimeter (GloMax®-Multi Detection System, Promega). Pure fluorescein diluted with potassium phosphate buffer (0–6 μg·ml^{-1}) was used as calibration solution, showing the following relation:

$$FC = 2 * 10^{-6} * F - 0.0044 (R^2 = 0.99)$$

where FC is the fluorescence concentration in μg·ml^{-1} and F is the fluorescence intensity in arbitrary units (AU).

2.7.2. Hydrolytic exoenzyme activities

The enzymatic activities present in the water extracts were determined using an APIZYM kit (BioMérieux Co, France) according to the supplier's instructions. The kit allows rapid monitoring of 19 hydrolytic enzymatic activities (proteases, glucosidases, lipases, etc.) without the need for purification. 65 μl of filtered supernatant used for the FDA analysis were placed in each well. The APIZYM gallery was covered and incubated at 37 °C for 4 h. After incubation, one drop of reagent ZYM A (BioMérieux Co, France) and one drop of reagent ZYM B (BioMérieux Co, France) were added to each

well. After 5 min incubation for the colour development, the gallery was exposed to a 500 W light source for about 10 s to eliminate colour from negative reactions. Finally, a score between 0 and 5 was assigned depending on the intensity of the residual colour by comparing each well with the control well, according to the supplier's instructions. This kit is reported in the literature to be able to identify qualitative hydrolytic exoenzyme activities in aerobic and anaerobic digestion studies (Charles et al., 2009; Tiquia et al., 2001; Xu et al., 2014b).

2.8. Statistical analyses

One factor analysis of variance (ANOVA) was performed to identify a difference between pre-treatment trials. In the cases that ANOVA tests were significant with p-value < 0.05, Fishers Least Significant Difference (LSD) tests were used to further investigate between groups.

3. Results and discussion

3.1. Impact of O_2 concentration and pre-treatment duration on further anaerobic biodegradability

3.1.1. Biodegradation kinetics of FW during aerobic pre-treatment

During the 1–4 days of aerobic pre-treatment, variations in the kinetics of gas production (CO_2, CH_4 and N_2O) were monitored on line to analyse aerobic and anaerobic metabolic activities. No production of CH_4 or N_2O was observed whatever the O_2 concentration in the entering gas flow, showing that methanogenesis or partial nitrification/denitrification activity did not take place during the pre-treatment (data not shown). Concerning CO_2, a first peak of production was observed in all conditions during the first day of pre-treatment (Fig. 1a) This CO_2 emission can result from biological activities such as respiration or fermentation, but also from physicochemical phenomena such as modification of the calco-carbonic balance. The fact that a similar amount of CO_2 was emitted during the first day without strong difference in the temperature increase (Fig. 1c), and whatever the O_2 level of the pre-treatment, suggests an abiotic process. This is confirmed by the CO_2:O_2 ratio (Fig. 1b) which was higher at 5% and 10%O_2 pre-treatments than at 21%O_2 during the first day, revealing the absence of a relation between O_2 consumption and CO_2 production, and which is also in accordance with the presence of that peak in anaerobic conditions (without O_2 consumption). Thus, the CO_2 peak observed during the first day of pre-treatment does not appear to be linked with the biodegradation process but rather with the degassing of carbonates present in FW. It results from the punctual disequilibrium induced between the gaseous phase and the solid aqueous phase by the establishment of the gas flow within the waste medium.

After one day of pre-treatment however, visible differences appeared between the different pre-treatment conditions. With 0%O_2, a minimum but constant production of CO_2 (~2.5 mmol CO_2·$kgTS^{-1}$·h^{-1}) was observed probably corresponding to microbial fermentation. In the low and fully aerobic conditions, CO_2 production rose, revealing a constant relation with O_2 consumption (Fig. 1a and b) and confirming an aerobic biological activity. These results are in agreement with those obtained by Botheju et al. (2010), who reported an increase in CO_2 generation with increasing O_2 load during an aerobic pre-treatment between 0% and 10%O_2 aeration on primary wastewater sludge. Moreover, our data (Fig. 1a) show that the production of CO_2 at 5%O_2 and 10%O_2 pre-treatment is similar but low compared to the one at 21%O_2. The total production of CO_2 at 4 days (T4) was 2255, 2435 and 4091 mmolCO_2·$kgTS^{-1}$ at 5%, 10% and 21%O_2 experiences respectively. These data show twice stronger biological activity, and thus biodegradation, at 21%O_2 compared to 5%O_2 and 10%O_2 pre-treatments. This observation is in line with the proposal of Haug (1993) stating that, for composting, O_2 concentrations lower than 10% in the air result in lower organic matter decomposition. Interestingly, even with experimental cells maintained in a thermostated bath, the increased biodegradation linked to biological respiration led to heat release and temperature increase (Fig. 1c), traducing the large extent of the aerobic exothermic reactions. Considering this high aerobic activity, organic matter degradation was quantified and BMP analyses were performed to assess the ability of these pre-treated FW to produce biogas in AD.

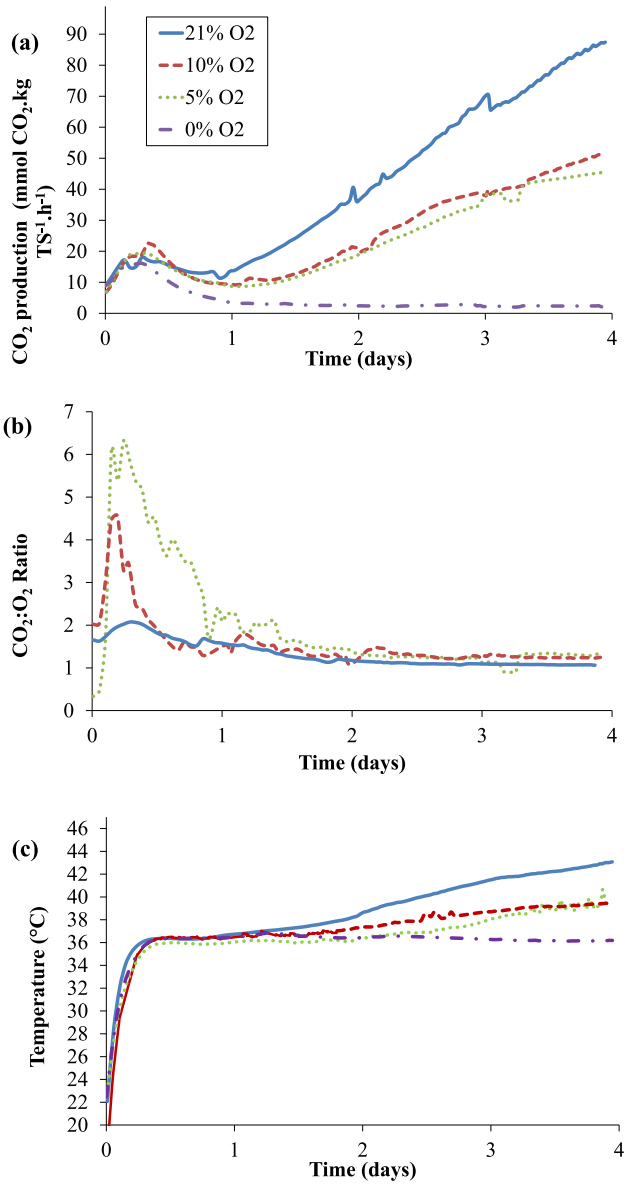


Fig. 1. (a) CO_2 production, (b) CO_2 production:O_2 consumption ratio, and (c) temperature variations within FW reactors subjected to gas flows with different oxygen contents (0% O_2 (purple, dashdotted line), 5% O_2 (green, dotted line), 10% O_2 (red, dashed line) and 21%O_2 (blue, solid line)).

3.1.2. Organic matter degradation after aerobic pre-treatment

VS and COD contents on FW after pre-treatment are shown on Table 2. No significant variation on VS was assessed at 5%, 10% and 21%O_2 for 1 and 2 days of pre-treatment and in all durations of anaerobic conditions (0%O_2) (p-value = 0.11). It is in agreement

Table 2
VS, COD, pH and BMP contents on raw FW (T0) and treated FW after different pre-treatment conditions (0%, 5%, 10% and 21%) and durations (1, 2, 3 and 4 days i.e. T1, T2, T3 and T4). * are used to indicate significant differences (p-value < 0.05) of pre-treated FW values with respect to raw FW (T0).

	Time	Pre-treatment condition			
		$0\%O_2$	$5\%O_2$	$10\%O_2$	$21\%O_2$
VS ($gVS{\cdot}gWW^{-1}$)	T0	256 ± 14	256 ± 14	256 ± 14	256 ± 14
	T1	281 ± 12	254 ± 8	271 ± 14	273 ± 9
	T2	262 ± 18	250 ± 5	260 ± 11	247 ± 6
	T3	260 ± 10	246 ± 5	240 ± 11	253 ± 3
	T4	248 ± 4	238 ± 5	233 ± 5*	224 ± 30*
COD ($gO_2{\cdot}kgWW^{-1}$)	T0	440 ± 13	440 ± 13	440 ± 13	440 ± 13
	T1	435 ± 30	408 ± 25	428 ± 24	441 ± 35
	T2	404 ± 30	402 ± 13	404 ± 16	392 ± 5
	T3	409 ± 23	362 ± 9*	439 ± 25	383 ± 97*
	T4	405 ± 22	336 ± 3*	400 ± 2	365 ± 15*
pH	T0	5.3	5.3	5.3	5.3
	T1	5.2	4.8	4.7	4.5
	T2	5.2	5.0	5.0	5.2
	T3	5.1	5.1	4.7	5.5
	T4	5.8	5.6	5.5	6.2*
BMP ($NLCH_4{\cdot}kgWW^{-1}$)	T0	127 ± 8	127 ± 8	127 ± 8	127 ± 8
	T2	128 ± 8	128 ± 1	132 ± 3	124 ± 4
	T4	115 ± 3*	117 ± 2*	121 ± 4	112 ± 7*
BMP ($NLCH_4{\cdot}kgVS^{-1}$)	T0	496 ± 32	496 ± 32	496 ± 32	496 ± 32
	T2	487 ± 29	512 ± 3	510 ± 13	501 ± 18
	T4	464 ± 13	492 ± 7	519 ± 17	500 ± 31

with Charles et al. (2009), who reported that two days of aerobic pre-treatment on OFMSW were optimal to avoid excessive VS degradation.

On longer pre-treatment duration, analyses showed significant reductions of VS (after three days of pre-treatment) and of COD (after two days of pre-treatment). The highest reduction of VS arose after four days of $21\%O_2$ pre-treatment, with a loss of 12% of the VS content of the raw FW. On the case of COD, the highest reductions were assessed after four days at $5\%O_2$ (p-value = 0.00) and $21\%O_2$ (p-value = 0.00) reaching 23% and 17% of COD loss compared to raw FW.

Given the different kinetics of biological activity observed during aerobic pre-treatments and the assessed organic matter degradations, differences were expected in the methane production of treated FW.

3.1.3. Anaerobic biodegradability of the pre-treated FW

Table 2 shows that methane yield measured based on wet weight, decreases with time during the pre-treatments. The difference was not statistically significant between the raw FW (T0) and the two-day pre-treated waste. However, a significant decrease (p-value = 0.01) of almost 7% was observed after 4 days of pre-treatment for all pre-treatment conditions (with the exception of $10\%O_2$). This reduction is consistent to the overall decrease on VS and COD assessed after 4 days of pre-treatment. These results confirm that an extended pre-treatment of FW is deleterious for further use of the waste in AD as also observed by Xu et al. (2014a).

Looking at the methane yield expressed on a VS basis, it appears that BMP also decreased by 6.5% with 4 days of a $0\%O_2$ pre-treatment (p-value = 0.04). It means that the most biodegradable organic matter was degraded during the pre-treatment leading to a less biodegradable residual organic matter for AD.

Interestingly, this reduction of the organic matter biodegradability was not observed for 4-day pre-treated FW with $5\%O_2$, $10\%O_2$ and $21\%O_2$ conditions, as their BMP expressed on a VS basis was not significantly different from the one of raw FW. These results suggest that the VS consumed by micro-organisms and mineralised into CO_2 during the aerobic pre-treatment, was balanced by changes in the quality of the residual VS content.

This hypothesis is reinforced when looking at the cumulative methane production curves. During the 10 first days of the BMP test, methane production is higher with pre-treated waste at 21% O_2 (378 $NLCH_4{\cdot}kgVS^{-1}$) (p-value = 0.01) than with raw FW (328 $NLCH_4{\cdot}kgVS^{-1}$) (Fig. 2c). With $5\%O_2$ pre-treated waste, methane production kinetics is comparable (after two days (T2) (314 $NLCH_4{\cdot}kgVS^{-1}$) (p-value = 0.11)) or slightly lower (after four days (T4) (285 $NLCH_4{\cdot}kgVS^{-1}$) (p-value = 0.01)) than with raw FW (T0) (328 $NLCH_4{\cdot}kgVS^{-1}$) (Fig. 2a). It can thus be assumed that the organic fractions of FW which were recalcitrant to anaerobic degradation when considering raw FW became more available and easily biodegradable thanks to the aerobic pre-treatment, especially with oxygen conditions higher than 10%.

To confirm this hypothesis and understand in depth the effect of the aerobic pre-treatment on the quality of the FW organic matter, its variations of composition, the biomass development and enzymatic activities were further studied.

3.2. Relation between variations in the biochemical characteristics of FW during aerobic pre-treatment and further anaerobic biodegradability

3.2.1. Variations in biochemical fractionation

To understand the variations in the quality of the FW organic matter during aerobic pre-treatment, a modified Van Soest fractionation was performed on all samples. This method allows discretising the total organic matter into several compartments of different putative biodegradability.

Looking at Fig. 3, it can be stated that the lignin-like fraction (LIG) remains globally unchanged whatever the pre-treatment applied. The other water non-soluble fractions (CEL, HEM and SOL_{NDF}) present some variations but no clear tendencies may be observed. However, a main difference appears in the reorganisation of the water soluble fractions (SOL_W > 1.5 kDa and SOL_W < 1.5 kDa). Indeed, the SOL_W > 1.5 kDa fraction seems to rise over time in all conditions of pre-treatment while at the same time the SOL_W < 1.5 kDa fraction was gradually consumed. This is especially significant for the aerobic pre-treatment with $21\%O_2$ from day 2 to 4. The increase of the SOL_W > 1.5 kDa fraction reveals

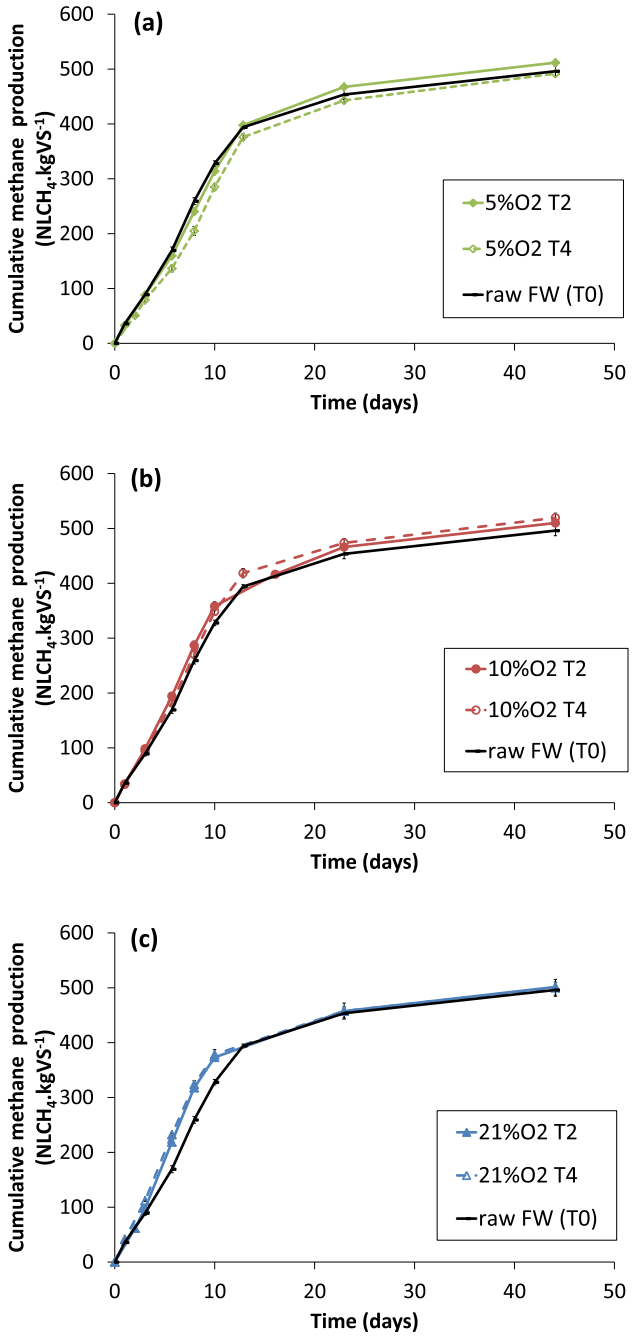


Fig. 2. Cumulative methane production of raw FW (T0) (-, black solid line) and after 2 days (T2) and 4 days (T4) of aerobic pre-treatment with oxygen concentrations of (a) 5%O_2 (◆, green), (b) 10%O_2 (●, red) and (c) 21%O_2 (▲, blue). Solid lines are used for T2, dashed lines are used for T4. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the solubilisation of slow biodegradable solid fractions of VS (like CEL, HEM and SOL_{NDF}) into more biodegradable molecules that compose the SOL_W > 1.5 kDa. Unfortunately, our experimental conditions did not allow observing clear variations of these slow biodegradable solid fractions.

The increase of the SOL_W > 1.5 kDa fraction could explain why the methane yield per kg of VS after aerobic pre-treatment remained unchanged. It also emphasises the importance of the composition of the water soluble fractions of the FW for further AD.

In addition, the losses of VS content after 4 days of pre-treatment (T4 in Table 2) can be mainly explained in all configurations by the decrease of this SOLW < 1.5 kDa fraction. It confirms that a short pre-treatment degrades the small water soluble molecules assumed to be the most biodegradable. In addition, biochemical fractionation of VS based on lipids, proteins and carbohydrates contents was also performed on all samples. However, no significant variations were assessed with our experimental conditions (data not shown).

3.2.2. Variations in simple carbohydrates and VFA concentrations

To investigate variations in the water soluble organic fractions (SOL_W > 1.5 kDa and SOL_W < 1.5 kDa) more in detail, their concentrations in simple carbohydrates (sucrose, glucose and fructose) and VFA were monitored. Fig. 4 shows that the simple carbohydrates content decreased regularly with time in both aerobic and anaerobic conditions. The unexpected strong decrease occurring on the third day of the 0% condition may be due to an undesired infiltration of air for about 12 h before stopping the cell. Aside from this uncontrolled result, the decrease in simple carbohydrates content was seen to be amplified by higher O_2 pre-treatment (Fig. 4a). It resulted in 80% removal of the total simple carbohydrates from the raw FW at high oxygenation (21%O_2, T4). In comparison, only 26% of the total simple carbohydrates were degraded in anaerobic conditions (0%O_2, T4). This simple carbohydrates diminution can be related to the diminution of the SOL_W < 1.5 kDa fraction (Fig. 3) and the high CO_2 production (Fig. 1a) at high O_2 concentration.

The VFA content was monitored in parallel showing an increase in their concentration with time in anaerobic conditions and limited oxygen condition (0% and 5%O_2), confirming microbial fermentations (Fig. 4b). In aerobic conditions, the accumulation of VFA was lower, with a slight decrease after 4 days at 10%O_2 (30% less than at 0%O_2 T4) and a marked decrease at 21%O_2 T4 (60% lower than 0%O_2 T4), which is clearly linked to the increase of pH from 5.3 to 6.2 at the latter condition as observed on Table 2. These results are consistent with previous reports of Rafieenia et al. (2017), using aeration pre-treatment in a two phase anaerobic digestion of FW, which observed a positive effect of pre-aeration on the reduction of VFA accumulation during the second stage of AD. The observed diminution in accumulation of VFA may result, either from the reduced availability of simple carbohydrates producing VFA as can be seen in Fig. 4a, or from a rate of VFA consumption higher than that of their production during high O_2 concentration pre-treatment.

Biochemical characteristics studied in this section give a better understanding of the BMP variations observed after aerobic pre-treatment. The stability of BMP can be attributed mainly to a positive effect of aerobic pre-treatment that produces a shift in the biodegradable molecules distribution: a diminution of very easily biodegradable molecules (SOL_W < 1.5 kDa fraction, simple carbohydrates), and an increase of the supply of soluble water molecules (SOL_W > 1.5 kDa fraction), recovered from the non-soluble fractions. These changes might help in case of subsequent AD, avoiding rapid simple carbohydrates consumption and therefore rapid VFA accumulation, making more rapidly available the less biodegradable molecules and conserving methane yield.

3.3. Relation between variations on biological characteristics of FW during aerobic pre-treatment and further anaerobic biodegradability

3.3.1. FW microbial community composition and dynamics

3.3.1.1. Microbial abundance. The number of bacteria and fungi in the raw FW was quantified by real-time qPCR at respectively $6.2 \cdot 10^{10}$ and $2.1 \cdot 10^{9}$ copies of rRNA gene per gram of raw FW (T0) total solids.

The concentration of bacteria remained relatively stable (variation of less than one log10) during the four days of pre-treatment. The number of fungi however was affected by the

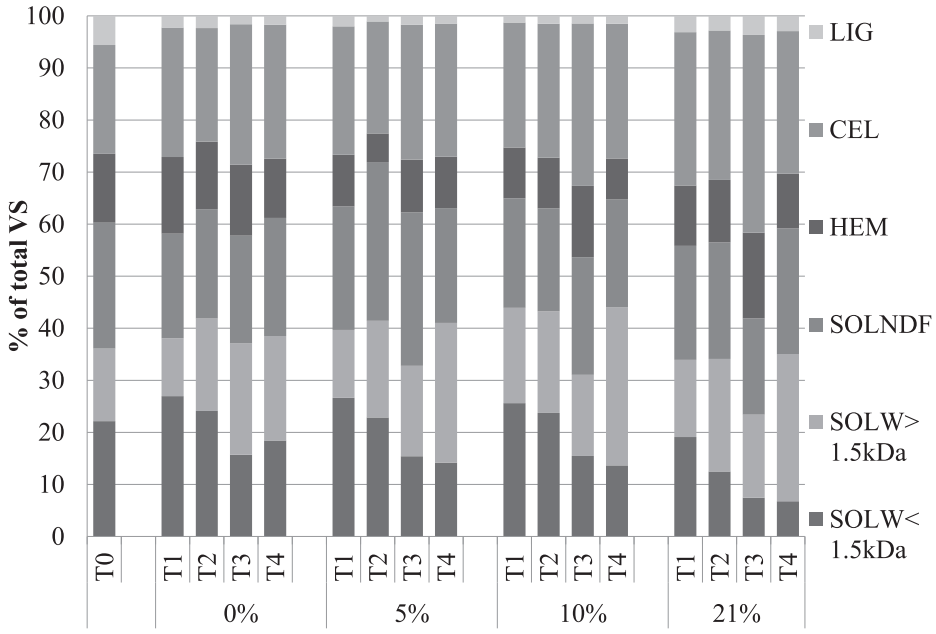


Fig. 3. Biochemical fractionation of VS of the raw FW (T0) and FW during aerobic pre-treatment at different oxygen contents (0%, 5%, 10% and 21% of O_2) and at different pre-treatment times (1, 2, 3 and 4 days i.e. T1, T2, T3 and T4). Organic matter (OM) fractions are: LIG: lignin-like OM, CEL: cellulose-like OM, HEM: hemicellulose-like OM, SOLNDF: OM soluble in neutral detergent, SOLW > 1.5 kDa: soluble in water OM molecules > 1.5 kDa and SOLW < 1.5 kDa: soluble in water OM molecules < 1.5 kDa.

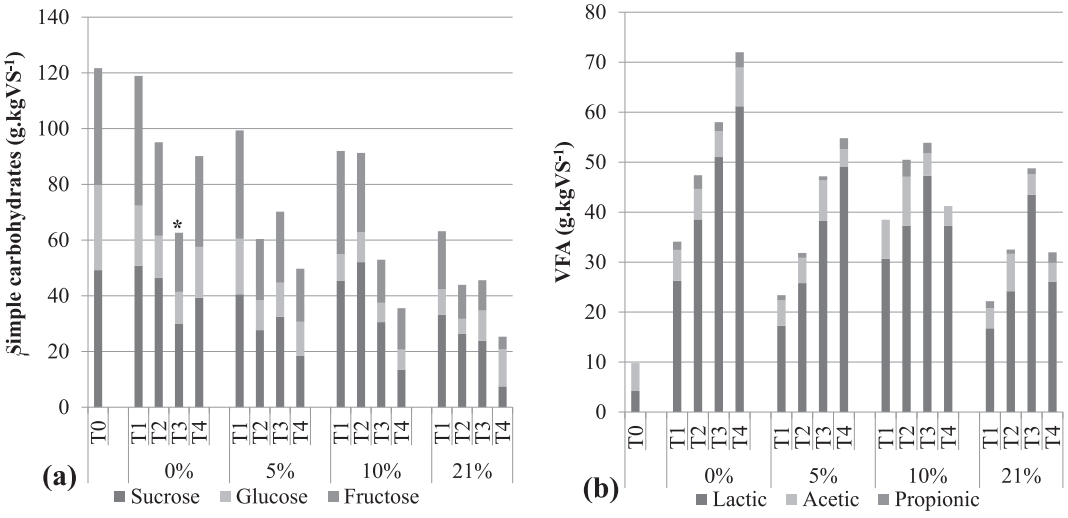


Fig. 4. Concentrations of (a) simple carbohydrates and (b) volatile fatty acids in raw FW VS (T0) and along aerobic pre-treatment (0%, 5%, 10% and 21% of O_2 content) with different duration (1, 2, 3 and 4 days i.e. T1, T2, T3 and T4). * Lower simple carbohydrates content at 3 days of 0% O_2 condition might result from an unexpected air infiltration.

O_2 concentration. After 2 days without oxygen (0%O_2), their number dropped by about one log10 to reach $3 \cdot 10^8$ rRNA genes·gTS^{-1} on day 2 and stabilized thereafter. In presence of oxygen, their number increased from the 2nd to the 4th day of pre-treatment and stabilized at values around 35, 19 and 4 times the ones observed on T0 for the 5%, 10% and 21% O_2 pre-treatment conditions. This result is in agreement with the aerobic metabolism of fungi but it is surprising that the strongest growth is seen for the lowest oxygen concentration. It could be explained by competition processes between fungi and bacteria for substrates as observed in other ecosystems (Mille-Lindblom et al., 2006).

3.3.1.2. Fungi identification. Fungi species were identified on day 4 of the 21%O_2 pre-treatment condition. Out of the 48 clones sequenced, 31 were identified as *Rhizopus oryzae* isolate A2 (acc. KM527239) and 12 as *Lichtheimia ramosa* (acc. LK023322), all with 99% of sequence similarity. These fungi belong to the Order *Mucorales* of the Class *Zygomycetes*, which are fast growing fungi. They are white filamentous fungi saprotrophic on plants, widespread in soil and found on a wide range of organic substrates including fruits and vegetables. *Rhizopus oryzae* secretes maltase and dextrinase and can produce alcohol and lactic acid through fermentation of starch under anaerobic conditions. Its fast growing and aerobic/anaerobic metabolism may have favoured its growth in pre-treatment conditions. Other sequences were identified as *Pichia* sp., *Candida* sp. and *Saccharomycetaceae*. Their known metabolic activities suggest that they may participate in exoenzymes production that will be studied in Section 3.3.2 (Meredith Blackwell, 2011).

Table 3
Number of sequences, OTUs and diversity indexes of the NGS dataset for each FW sample. As the values fluctuate randomly from time T1 to T4 in each %O_2 pre-treatment condition, the table presents the minimum and maximum values observed within each condition.

	Raw FW	Pre-treated FW			
		0% O_2 (min-max)	5% O_2 (min-max)	10% O_2 (min-max)	21% O_2 (min-max)
Number of raw sequences	28,133	20,796–29,111	6452–62,097	24,162–32,293	22,491–83,525
Number of sequences after treatment	5094	5081–15,644	1394–9587	6059–17,856	5071–16,019
Number of OTUs (richness)	64	32–70	38–47	46–96	40–65
Diversity coverage (%)	98.5	98.4–99.5	98.5–99.5	97.1–98.9	98–99.2
Evenness	0.67	0.63–0.68	0.59–0.67	0.58–0.69	0.64–0.73
Simpson	0.89	0.84–0.89	0.81–0.87	0.84–0.91	0.86–0.91
Shannon-Weaver	4.01	3.24–4.01	3.07–3.58	3.25–4.49	3.43–4.16

3.3.1.3. Bacterial diversity analysis. The high throughput DNA sequencing produced a mean of 10,105 ± 4937 16S rDNA sequences per sample retrieved every day for all %O_2 pre-treatment conditions except for the 5%O_2 pre-treatment sample of day 4 (Table 3). This latter sample repeatedly gave a low number of correct sequences (1394) for unknown reasons. As observed in Fisgativa et al. (2017) and David et al. (2014), the data from the raw FW (T0) were strongly polluted by DNA sequences identified as chloroplasts (*Cyanobacteria*/*Streptophyta* group) originating from FW vegetables (data not shown). These sequences represented 64% of the FW T0 total sequences but their number dropped notably after one day of pre-treatment (10–15% at T1 and only a few percent at T2 to T4). These sequences were removed from further analysis without altering the representativeness of the data. Actually, once these artefactual sequences were removed, the mean coverage for all DNA sequence libraries was still very good at 98.7% with a minimum of 97.1% (Table 3).

The number of dominant OTUs (molecular representation of species) per sample was relatively low and stable with time. It averaged 64 OTUs for the raw FW (T0) and 48 OTUs ± 17 for T1, 43 OTUs ± 10 for T2, 71 ± 24 for T3 and 51 ± 11 for T4 at the aeration conditions of0%, 5%, 10% and 21%O_2 respectively. The α-diversity indexes (Simpson and Shannon-Weaver that represent bacterial diversity in each sample) were relatively stable with time and were within the range observed previously on other FW by Fisgativa et al. (2017). All these data suggest a low and stable bacterial diversity, concordant with stability of the total number of bacteria observed above by qPCR.

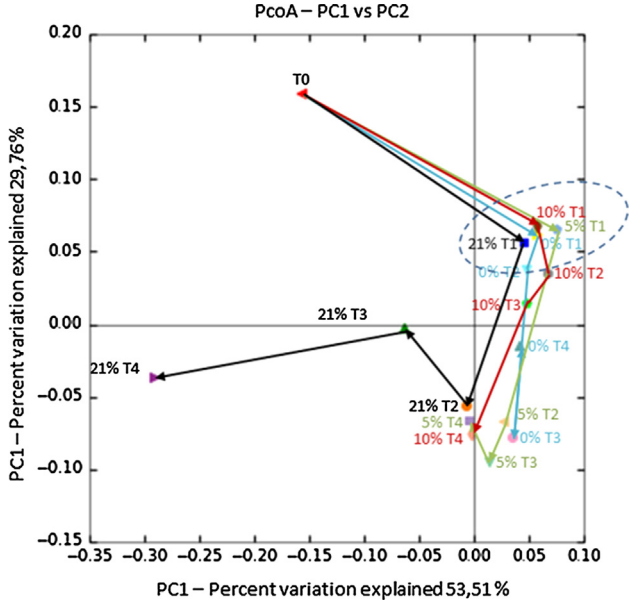


Fig. 5. Principal coordinate plots of the distances between the microbial community structures of all samples with T0 to T4 being the duration of pre-treatment and 0%, 5%, 10% and 21% the O_2 concentration of pre-treatment.

Statistical analysis of the β-diversity indexes of all samples allowed the comparison of the FW bacterial community composition during pre-treatments (Fig. 5). It shows that the FW bacterial community composition changed considerably but similarly during the first day of pre-treatment (from T0 to T1) whatever the O_2 concentration. This suggests the existence of one or more parameters having a stronger selective impact on bacterial communities than oxygen. It could be the temperature applied to the cells since this period corresponds to the increase in temperature from 20 °C to 36 °C as shown in Fig. 1c. After one day of pre-treatment, the FW bacterial communities evolved in the same way except for the 21%O_2 pre-treatment that changed more rapidly and deeply than for other pre-treatment conditions. This suggests that the most important parameter after one day of pre-treatment is the O_2 concentration and that a stronger impact is seen for the highest concentration of 21%O_2.

3.3.1.4. Changes in the FW bacterial community's composition during pre-treatments. When the DNA sequences were analysed at the species level, no *Archaea* was observed and less than 4% of the total sequences belonged to low abundant OTUs (i.e. OTUs containing less than 1% of the total sequences). Actually, between 80% and 100% of the OTUs having less than 1% of the total sequences belonged to the *Firmicutes* and the *Proteobacteria* phyla.

Fig. 6 shows the impact of the 0%O_2 and 21%O_2 pre-treatments on the FW microbial community composition. In the raw FW (Fig. 6 T0), the *Proteobacteria* and *Firmicutes* phyla were almost equally abundant making up to respectively 51% and 46% of the total sequences. The *Proteobacteria* were essentially members of the *Pseudomonadaceae* and *Enterobacteriaceae* families and were dominated by a *Pseudomonas* specie that represented 27% of the total sequences. The *Firmicutes* were almost totally composed of a *Leuconostoc* species that made up to 38% of the total sequences. *Pseudomonas* is a strict aerobic bacterium widespread in the environment and with hydrolytic activities. Its presence is expected in FW. *Leuconostoc* is a facultative anaerobe bacterium that ferments carbohydrates to produce lactic acid. Its presence in the raw FW at such high concentration suggests that the collected waste was already in a fermentation process during storage stage and before freezing.

During pre-treatments, the FW microbial community composition changed differently depending on the presence or absence of O_2 (Fig. 6). For the 0%O_2 pre-treatment (Fig. 6a), the proportion of *Proteobacteria* decreased dramatically in one day (T0 to T1) and remained very low, ending by representing less than 3% of the total sequences. In the meantime, the *Firmicutes* increased up to 90% of the total sequences on T1 and 97% on T4. This increase was accompanied by a change of the dominant species. The *Leuconostoc* identified in the raw FW was slowly replaced by two *Lactobacillus* species. This shift from *Leuconostoc* to *Lactobacillus* may

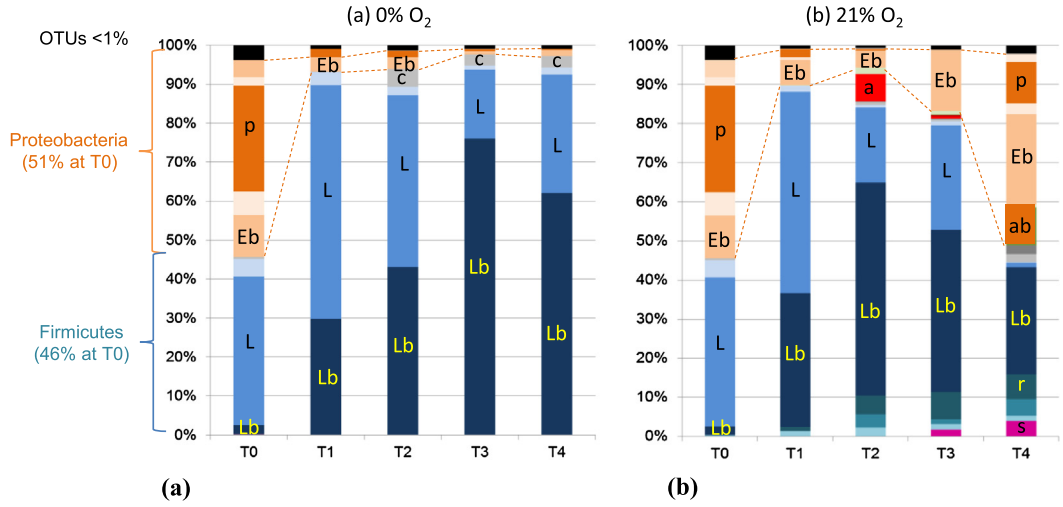


Fig. 6. Evolution of the main bacterial groups during the four days of: (a) 0%O_2 pre-treatment and (b) 21%O_2 pre-treatment. The *Proteobacteria* are represented by various orange colours, the *Firmicutes* by various blue and green colours. The identified genera were: P. *Pseudomonas*; L. *Leuconostoc*; Lb. *Lactobacillus*; c. *Clostridiales*; a. *Acetobacter*; Eb. *Enterobacteriaceae*; ab. *Achromobacter*; r. *Rummeliibacillus*; s. *Sphingobacterium*.

be explained by the elevation of FW temperature from 20 °C to 36 °C (Fig. 1c). The optimum temperature of *Leuconostoc* is 20–30 °C while the one of *Lactobacillus* is between 30 °C and 40 °C. Since both genera are facultative anaerobe bacteria that ferment carbohydrates to produce lactic acid (Castenholz et al., 2001), their growth explains the degradation of simple carbohydrates and high lactic acid proportion in VFA seen in Fig. 4. Interestingly, members of *Clostridiales* were identified from days 2–4 of the 0%O_2 pre-treatment confirming the strict anaerobic conditions of the waste.

For the 21%O_2 pre-treatment (Fig. 6b), the first day of pre-treatment (T1) also induced a strong decrease of the *Proteobacteria* down to less than 10% of the total sequences and an increase of the *Firmicutes* up to 90% of the total sequences. However, on the contrary of the 0%O_2 pre-treatment, the proportion of *Proteobacteria* increased again after 2, 3 and 4 days of pre-treatment to reach respectively 14%, 18% and 49% of the total sequences at T4. The dominant *Pseudomonas* observed in the raw FW (27% of the total sequences at T0) was rapidly replaced by environmental species of *Enterobacteriaceae* (22% of the sequences on T4) like *Klebsiella*, *Enterobacter* or *Serratia*. In the meantime, the same shift of dominant *Firmicutes* species from *Leuconostoc* to *Lactobacillus* was observed. *Pseudomonas* is a strict aerobic bacterium whereas most *Enterobacteriaceae* can grow in absence of O_2 by fermenting simple carbohydrates and produce lactic acid, alcohols and various other end products. These populations' shifts suggest that the microbial community was limited in O_2 during the first day of pre-treatment, even at 21%O_2, transiently limiting the growth of aerobic *Proteobacteria* and favouring the fermentative lactic acid bacteria *Leuconostoc* and *Lactobacillus*. This transient limitation may result from the uptake of all the O_2 by the FW microorganisms whose growth was stimulated by both the aeration and cell warming. The appearance of *Acetobacter* on days T2 to T4 can be related to the presence and consumption of VFAs in the FW observed in Fig. 4b. *Acetobacter* is a strict aerobic non-fermentative *α-Proteobacteria* that mineralises acetic and lactic acids into CO_2 and H_2O. It can grow at pH as low as 5.4. Its appearance after two days of pre-treatment at 5%, 10% and 21%O_2 suggests that it may counteract the action of the lactic acid bacteria.

3.3.2. Variations in enzymes content of FW

3.3.2.1. Total exoenzymes content. The FDA method was used to determine the total hydrolytic exoenzymes content of FW during pre-treatment as described in Section 2.7. The hydrolytic exoenzymes are known to be involved in the breakdown of large macromolecules, enhancing the proportion of low molecular weight organic molecules on the waste (Zhang et al., 2007). As shown in Table 4, a progressive increase in the production of exoenzymes was clearly observed with 21%O_2, with two or three fold higher values after 3 and 4 days of pre-treatment than in the other aerobic conditions. More irregular production was observed with 5% and 10%O_2 showing that these conditions limit the production of exoenzymes. At 0%O_2, the exoenzymes content decreased and was 8 and 15 times lower than the one with 21%O_2 after 3 and 4 days of pre-treatment respectively. These results show that the exoenzymes are mainly produced under full aerobic conditions, which can be related to the diversification of the microbial community in the waste. Higher exoenzymes content may also be related to the hypothesis that the increase in the SOL_W > 1.5 kDa fraction (Fig. 3) is caused by solubilisation of OM from the NSOL fraction thanks to the increased activity of hydrolytic exoenzymes, as also observed by Charles et al. (2009) on OFMSW.

3.3.2.2. Qualitative activity of exoenzymes. Table 5 shows the range of hydrolytic enzymes present in raw FW (T0) and during the

Table 4
Exoenzymes quantification on raw FW (T0) and after different pre-treatment conditions (0%, 5%, 10% and 21%) and durations (1, 2, 3 and 4 days i.e. T1, T2, T3 and T4).

	Time	Pre-treatment condition			
		0%O_2	5% O_2	10% O_2	21% O_2
Exoenzymes production ($\mu gF \cdot gTS^{-1} \cdot h^{-1}$)	T0	92 ± 2	92 ± 2	92 ± 2	92 ± 2
	T1	113 ± 4	127 ± 1	198 ± 2	45 ± 1
	T2	101 ± 1	214 ± 1	112 ± 1	126 ± 1
	T3	51 ± 0	90 ± 1	141 ± 2	423 ± 3
	T4	50 ± 1	210 ± 1	305 ± 1	784 ± 11

Table 5
Qualitative activity of exoenzymes extracted from raw FW (T0) and FW pre-treated under different oxygen conditions (0%, 5%, 10% and 21% O_2) after one (T1) to four (T4) days of pre-treatment.

Enzymatic group	Enzyme type	Raw FW	0% O_2				5% O_2				10% O_2				21% O_2			
		T0	T1	T2	T3	T4	T1	T2	T3	T4	T1	T2	T3	T4	T1	T2	T3	T4
Phosphatases	Alkaline phosphatase	4	1	1	2	1	3	3	3	4	4	4	4	5	0	3	4	5
	Acid phosphatase	5	5	4	3	3	5	5	5	5	5	5	5	5	5	5	5	5
	Naphthol-AS-BI-phosphohydrolase	5	5	2	2	1	5	4	3	3	5	5	4	5	5	5	5	5
Lipase	Esterase (C 4)	2	5	3	1	0	1	2	2	3	2	2	2	2	2	2	2	3
	Esterase Lipase (C 8)	0	1	1	0	0	0	0	0	1	0	0	0	0	1	1	1	1.5
	Lipase (C 14)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.5
Amino-Peptidase	Leucine arylamidase	5	0	0	0	0	0	3	2	5	1	2	3	4	0	3	5	5
	Valine arylamidase	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Cystine arylamidase	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Proteases	Trypsin	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	α-chymotrypsin	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2
Glycosyl-hydrolases	α-galactosidase	0	2	2	1	0	2	2	2	2	1	1	3	3	2	0	2	2
	ß-galactosidase	2	4	4	3	3	4	4	5	5	4	5	5	5	4	4	4	4
	ß-glucuronidase	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
	α-glucosidase	1	3	3	0	0	2	2	3	4	2	2	2	2	3	2	4	4
	ß-glucosidase	1	3	3	0	0	3	2	4	4	2	3	3	3	2	2	3	3
	N-acetyl-ß-glucosaminidase	4	4	4	2	2	4	3	4	4	3	3	3	3	5	2	4	4
	α-mannosidase	0	1	1	0	0	0	0	0	0	1	1	0	1	2	0	1	1
	α-fucosidase	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Total activity		31	34	29	14	10	29	30	33	40	30	33	34	38	31	29	40	46

different pre-treatments (0–21%O_2 for one to four days). The main enzyme categories identified were phosphatases and glycosidases. Phosphatases are linked to the global cellular metabolism and alkaline phosphatases are involved in respiratory activity (Liberti et al., 2013). These enzymatic activities were thus logically observed in raw FW and during aerobic conditions (10% and 21% O_2 and to a lesser extent in 5%O_2). They diminished in anaerobic conditions (0%O_2). Few lipid and protein degradation activities were detected. These results are coherent with observations of Martínez-Valdez et al. (2016) in which low protease activity was assessed from OFMSW aerobically pre-treated up to 4 days. In our case, only a leucine arylamidase activity which is involved in the first steps of extracellular protein degradation (Muller et al., 2004), was observed in presence of O_2. Glucosidases are involved in the biodegradation of carbohydrates, especially cellulose (Teeri, 1997). These activities were low in the raw FW but ß-galactosidase, α- and ß- glucosidases and N-acetyl-ß-glucosaminidase increased with aerobic pre-treatments. This increase of glucosidases may be also compared to the increase of amylase from 0 to 6 days of aerobic pre-treatment performed by Martínez-Valdez et al. (2016). The transient increase on days T1 and T2 at 0%O_2 could result from slight introduction of oxygen when filling the cell with FW.

It can be seen that aerobic pre-treatment induces secretion of an interesting hydrolytic cocktail of enzymes in FW. A higher concentration and diversification of enzymes (higher content of protein degrading enzymes (Phosphatases and Amino-peptidases) and carbohydrates degrading enzymes (ß-galactosidase, α- and ß- glucosidases and N-acetyl-ß-glucosaminidase) were mainly stated at 21%O_2 conditions. This enriched enzymatic cocktail may help enhance hydrolysis of proteins and carbohydrates at subsequent AD, especially in case of process with low inoculum to substrate ratio like dry AD. Indeed in this latter condition the own biological composition (bacterial and enzymatic) of the FW may strongly impact anaerobic digestion.

4. Conclusion

In this work, four O_2 concentrations (0%, 5%, 10% and 21%) were tested as pre-treatment of FW with 4 different durations (1, 2, 3 and 4 days). The results suggested that aerobic pre-treatment of FW could be used before anaerobic digestion to improve waste biodegradability and also to limit the detrimental effect of acidification. Pre-treatments with limited oxygen (5% and 10%O_2) are less efficient and also more difficult to implement. Prolonging pre-treatment at 21%O_2 up to 4 days reduces significantly the FW methane potential, so the cost-benefit of its implementation needs to be evaluated on a case by case basis. A short aerobic pre-treatment (up to 4 days) strongly modifies the microbial community of FW promoting the presence of fungi and VFA consuming bacteria like *Acetobacter* which may contribute to limit VFA accumulation. It enhances the solubilisation of the waste organic matter and thus allows a preservation of the waste methane potential. To conclude, since VFA accumulation may lead to real instabilities and operational difficulties during FW anaerobic digestion, a short aerobic pre-treatment appears as an interesting and robust option to be tested to facilitate FW AD management, especially considering dry processes with low inoculum to substrate ratio.

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